

AN IMPROVED KISSING NUMBER IN DIMENSION 27

BEN LINDOW

ABSTRACT. The best known lower bounds for the kissing number in dimensions 25 through 31 all come from a single construction of Cohn, Jiao, Kumar and Torquato, which lifts subsets of the minimal vectors of the Leech lattice into $d = n - 24$ extra dimensions along a partition of a kissing configuration in $\mathbb{R}^d$. For $n = 27$ that configuration is the cuboctahedron, and both Kallal–Kan–Wang and Ma et al. partition it into two equilateral triangles and three antipodal pairs, of total weight 7. We observe that the cuboctahedron instead partitions into *four* pairwise disjoint equilateral triangles, of weight 8. Feeding this partition into the same template, with four of Ma et al.’s 496-element subsets of the Leech minimal vectors and their twelve extra spheres, gives

$$K(27) \geq 12 + 196560 + 8 \cdot 496 = 200540,$$

improving the previous record 200044. The resulting configuration of 200540 unit spheres is exhibited explicitly and verified in exact arithmetic. We also show that the partitions used in dimensions 25, 26 and 28 through 31 are optimal, so within this template no analogous improvement is available there.

1. INTRODUCTION

The kissing number $K(n)$ is the largest number of pairwise non-overlapping unit balls in $\mathbb{R}^n$ touching a fixed central unit ball; equivalently, the largest size of a set of unit vectors in $\mathbb{R}^n$ with pairwise inner products at most $1/2$. Exact values are known only for $n \in \{1, 2, 3, 4, 8, 24\}$ [10, 9].

In dimensions 25 through 31 every record lower bound to date comes from one construction, Theorem 7.5 of Cohn, Jiao, Kumar and Torquato [1]: take the 196 560 minimal vectors of the Leech lattice Λ_{24} , select mutually disjoint subsets $S_1, \dots, S_k$ in which no two vectors meet at exactly 60° , and lift each S_i into the $d = n - 24$ extra dimensions along a group T_i of directions drawn from a kissing configuration in $\mathbb{R}^d$. Kallal, Kan and Wang [5] found subsets of size 488 by simulated annealing; Ma et al. [7] reached 496 with a reinforcement-learning system and also sharpened the template, by adding $K(d)$ further spheres lying entirely in the extra dimensions and by partitioning the $\mathbb{R}^d$ configuration into more equilateral triangles. Their values are the ones currently listed in Cohn’s table [2]. (They are nowhere near the corresponding upper bounds — 522 212 in dimension 27 [4] — so the interest lies in the constructions.)

The count is linear both in the sizes $|S_i|$ and in the *weight*

$$w = \sum_i (|T_i| - 1) \tag{1}$$

of the partition of the $\mathbb{R}^d$ configuration. This note concerns the second factor, in one dimension only. For $n = 27$ we have $d = 3$, and the $\mathbb{R}^3$ configuration used is the cuboctahedron — the 12 vertices of the $A_3 = D_3$ root system, of optimal size since $K(3) = 12$. Both prior papers partition it into two equilateral triangles and three antipodal pairs, of weight $2 + 2 + 1 + 1 + 1 = 7$. Our observation is purely about $\mathbb{R}^3$: *the cuboctahedron is the disjoint union of four equilateral triangles*, of weight 8. This buys one further copy of $|S| = 496$.

Theorem 1. $K(27) \geq 200540$.

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The previous record is 200044 [7, 2]. The configuration realising Theorem 1 is written out in the accompanying data files and verified exactly; see Section 5.

What is new, and what is reused. New here is only Lemma 3, a statement about twelve points in $\mathbb{R}^3$, together with the optimality analysis of Section 4. Everything else is prior work: the lifting template and all its inequalities are Theorem 7.5 of [1]; the extra spheres and the emphasis on triangles are due to [7]; the four 496-element subsets in our certificate are, verbatim, four of the five subsets that [7] themselves use in dimension 27 (recovered from their published configurations and converted to our coordinate labelling), and the twelve extra spheres are theirs unchanged. We claim no new independent set in the Leech conflict graph: 496 remains the record there, and [7] conjecture it is optimal. The improved partition in fact uses *fewer* of the 496-sets than the old one — four instead of five.

2. THE LIFTING TEMPLATE

We use the norm-4 normalisation of [1, 5]: a kissing configuration in $\mathbb{R}^m$ is a set of vectors of squared norm 4 with all pairwise inner products at most 2. Let $C \subset \mathbb{R}^{24}$ be the 196 560 minimal vectors of the Leech lattice [6, 3], scaled so that $|x|^2 = 4$; for distinct $x, x' \in C$ one has $\langle x, x' \rangle \in \{-4, -2, -1, 0, 1, 2\}$.

Theorem 2 ([1, Thm. 7.5], with the extra spheres of [7]). *Let $T \subset \mathbb{R}^d$ be unit vectors with $\langle y, y' \rangle \leq 1/2$ for distinct $y, y' \in T$, and let $T_1, \dots, T_k \subseteq T$ be pairwise disjoint with $\langle y, y' \rangle \leq -1/2$ for distinct y, y' in the same T_i . Let $S_1, \dots, S_k \subseteq C$ be pairwise disjoint with $\langle x, x' \rangle \leq 1$ for distinct $x, x' \in S_i$, and put $U = \bigcup_i S_i$. Let $E \subset \mathbb{R}^d$ be unit vectors with $\langle y', y'' \rangle \leq 1/2$ for distinct $y', y'' \in E$ and $\langle y', y \rangle \leq \sqrt{3}/2$ for all $y' \in E, y \in T$. Then*

$$X = \{(x, 0) : x \in C \setminus U\} \cup \left\{ \left(\sqrt{\frac{2}{3}}x, \sqrt{\frac{4}{3}}y \right) : x \in S_i, y \in T_i \right\} \cup \{(0, 2y') : y' \in E\}$$

is a kissing configuration in $\mathbb{R}^{24+d}$, so

$$K(24 + d) \geq |E| + 196560 + \sum_{i=1}^k (|T_i| - 1)|S_i|. \quad (2)$$

Proof sketch. Every listed vector has squared norm 4 ($\frac{2}{3} \cdot 4 + \frac{4}{3} = 4$ for a lifted vector). Writing $p = \langle x, x' \rangle$ and $c = \langle y, y' \rangle$, the inner products fall into eight classes, each reducing to a hypothesis:

pair class	inner product	bounded by 2 because
equatorial–equatorial	p	$ x - x' ^2 \geq 4$, so $p \leq 2$
equatorial–lifted	$\sqrt{2/3}p$	$\sqrt{2/3} \cdot 2 < 2$
lifted–lifted, same x	$\frac{8}{3}p + \frac{4}{3}c$	$c \leq -\frac{1}{2}$
lifted–lifted, same $S_i, y = y'$	$\frac{8}{3}p + \frac{4}{3}c$	$p \leq 1$
lifted–lifted, same $S_i, y \neq y'$	$\frac{8}{3}p + \frac{4}{3}c$	$p \leq 4, c \leq -\frac{1}{2}$
lifted–lifted, $S_i \neq S_j$	$\frac{8}{3}p + \frac{4}{3}c$	$p \leq 2, c \leq \frac{1}{2}$
extra–equatorial	0	
extra–lifted	$\frac{4}{\sqrt{3}}c$	$c \leq \frac{\sqrt{3}}{2}$
extra–extra	$4c$	$c \leq \frac{1}{2}$

The sixth line is where disjointness of the S_i is used: distinct sets force $x \neq x'$, hence $p \leq 2$, and the two bounds meet exactly at 2. Finally $|X| = (196560 - |U|) + \sum_i |T_i||S_i| + |E|$, which is (2). $\square$

A group T_i has $|T_i| \leq 3$: for four unit vectors with pairwise inner products $\leq -1/2$, $0 \leq |v_1 + \dots + v_4|^2 = 4 + 2 \sum_{i < j} \langle v_i, v_j \rangle \leq 4 - 6 < 0$. So a group is an antipodal pair, a pair at 120° , or an equilateral triangle through the centre, of weight 1, 1, 2 respectively. Triangles are twice as

good as pairs, which is why their number matters. The forms of [7] are exactly (2) with $|S_i| = 496$ and weights $w = 1, 4, 7, 16, 26, 48, 84$ for $n = 25, \dots, 31$.

3. THE CUBOCTAHEDRON SPLITS INTO FOUR TRIANGLES

Realise the cuboctahedron as the A_3 root system

$$A_3 = \{e_i - e_j : 1 \leq i \neq j \leq 4\} \subset \left\{v \in \mathbb{R}^4 : \sum_m v_m = 0\right\},$$

of common squared norm 2; the unit vectors of T are these divided by $\sqrt{2}$. Distinct roots have inner product in $\{-2, -1, 0, 1\}$, so all pairwise cosines lie in $\{-1, -\frac{1}{2}, 0, \frac{1}{2}\}$, as a kissing configuration requires. Three roots form an equilateral triangle (a valid group of size 3) if and only if their pairwise inner products are all -1 , equivalently if and only if they sum to 0.

Lemma 3. *The cuboctahedron is the disjoint union of four equilateral triangles. Explicitly, writing $(i j k)$ for $\{e_i - e_j, e_j - e_k, e_k - e_i\}$, the four triples*

$$(1 2 3), \quad (1 3 4), \quad (1 4 2), \quad (2 4 3)$$

are pairwise disjoint equilateral triangles whose union is all of A_3 .

Proof. Each triple sums to zero and has pairwise inner products $\langle e_i - e_j, e_j - e_k \rangle = -1$, so each is an equilateral triangle. Identify the root $e_i - e_j$ with the arc $i \rightarrow j$ of the complete directed graph on $\{1, 2, 3, 4\}$, which has 12 arcs; then $(i j k)$ is the directed 3-cycle $i \rightarrow j \rightarrow k \rightarrow i$. The twelve arcs of the four cycles listed are

$$\begin{array}{lll} 1 \rightarrow 2, & 2 \rightarrow 3, & 3 \rightarrow 1; \\ 1 \rightarrow 4, & 4 \rightarrow 2, & 2 \rightarrow 1; \end{array} \quad \begin{array}{lll} 1 \rightarrow 3, & 3 \rightarrow 4, & 4 \rightarrow 1; \\ 2 \rightarrow 4, & 4 \rightarrow 3, & 3 \rightarrow 2, \end{array}$$

pairwise distinct and hence exhausting all twelve arcs. $\square$

Remark 4. A decomposition of the complete symmetric digraph on v points into directed triangles is a Mendelsohn triple system of order v [8]; these exist exactly for $v \equiv 0, 1 \pmod{3}$, $v \neq 6$. Lemma 3 is the case $v = 4$. In the D_3 model $\{(\pm 1, \pm 1, 0)\}$ used in the data files, the four triangles are

$$\begin{aligned} T_1 &= \{(-1, -1, 0), (0, 1, -1), (1, 0, 1)\}, & T_2 &= \{(-1, 0, -1), (0, 1, 1), (1, -1, 0)\}, \\ T_3 &= \{(-1, 0, 1), (0, -1, -1), (1, 1, 0)\}, & T_4 &= \{(-1, 1, 0), (0, -1, 1), (1, 0, -1)\}, \end{aligned}$$

each summing to 0; the isometry $A_3 \rightarrow D_3$ carrying one description to the other is determined by $e_1 - e_2 \mapsto (-1, -1, 0)$, $e_2 - e_3 \mapsto (0, 1, -1)$, $e_3 - e_4 \mapsto (1, -1, 0)$.

Proof of Theorem 1. Apply Theorem 2 with $d = 3$, $k = 4$, T the cuboctahedron and $T_1, \dots, T_4$ the four triangles of Lemma 3. For $S_1, \dots, S_4$ take four pairwise disjoint 496-element subsets of C with no two vectors at 60° : the dimension-31 configuration published by [7] decomposes exactly into forty-two such sets, of which the first five are the ones they use in dimension 27; we take four of those five. For E take [7]'s twelve dimension-27 extra spheres, namely

$$E = \{(0, \pm 1, 0), (0, 0, \pm 1)\} \cup \left\{\left(\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{2}, \pm \frac{1}{2}\right)\right\},$$

the cuboctahedron rotated by 45° in the plane of $(1, 1, 0)$ and $(1, -1, 0)$; its pairwise cosines are at most $1/2$, and its cosines against T are at most $(2 + \sqrt{2})/4 = 0.85355\dots < \sqrt{3}/2$. Then (2) gives $K(27) \geq 12 + 196560 + 4 \cdot 2 \cdot 496 = 200540$. $\square$

TABLE 1. Bounds from (2) with $|S_i| = 496$. “used” is the weight of the partition of [7]; “max” is the ceiling of Proposition 5. Only $n = 27$ changes.

n	d	T	$K(d)$	$ E $	used	max	previous	this note
25	1	A_1	2	0	1	1	197 056	197 056
26	2	A_2	6	6	4	4	198 550	198 550
27	3	$A_3 = D_3$	12	12	7	8	200 044	200 540
28	4	D_4	24	24	16	16	204 520	204 520
29	5	D_5	40	40	26	26	209 496	209 496
30	6	E_6	72	72	48	48	220 440	220 440
31	7	E_7	126	126	84	84	238 350	238 350

4. THE OTHER DIMENSIONS

The same slack does not exist elsewhere. The weight is bounded by a counting argument using nothing but $|T_i| \leq 3$.

Proposition 5. *Let T be any set of K unit vectors in $\mathbb{R}^d$ and let $T_1, \dots, T_k \subseteq T$ be pairwise disjoint with $\langle y, y' \rangle \leq -1/2$ for distinct y, y' in the same subset. Then*

$$w = \sum_i (|T_i| - 1) \leq \left\lfloor \frac{2K}{3} \right\rfloor = 2 \left\lfloor \frac{K}{3} \right\rfloor + [K \equiv 2 \pmod{3}].$$

Proof. Groups of size ≤ 1 contribute nothing; discard them. Say a groups have size 3 and b have size 2 (no larger, as shown above). Disjointness gives $3a + 2b \leq K$, and $w = 2a + b$, so $3w = 6a + 3b \leq 2(3a + 2b) \leq 2K$. For the second form write $K = 3m + r$, $r \in \{0, 1, 2\}$; then $\lfloor 2K/3 \rfloor = 2m + [r = 2]$. $\square$

Proposition 5 uses no geometry beyond the group condition: it applies to *any* K -point configuration, lattice or not. Since E is itself a kissing configuration in $\mathbb{R}^d$ we also have $|E| \leq K(d)$, so the whole $\mathbb{R}^d$ side is bounded by data depending only on $K(d)$:

$$|E| + 196560 + w \cdot |S| \leq K(d) + 196560 + \left\lfloor \frac{2K(d)}{3} \right\rfloor |S| \quad (3)$$

when the sets share a common size $|S|$. This ceiling is attained for $2 \leq d \leq 7$, though not for $d = 1$, where E must be empty: every unit vector of $\mathbb{R}^1$ is ± 1 and has cosine 1 with some element of $T = \{\pm 1\}$, exceeding $\sqrt{3}/2$. Thus $K(25) \geq 196560 + |S|$, and $|S|$ is the entire game in dimension 25.

Table 1 compares, in each dimension, the weight used by [7] with the ceiling of Proposition 5; the ceiling is attained by an explicit partition of the standard root-system model in every case, and only $d = 3$ was previously short of it. Two comments.

(a) *Dimensions 25–28 are now closed within the template.* For $d \leq 4$ the kissing number is known exactly (2, 6, 12, 24), so Proposition 5 bounds the weight absolutely: no configuration in $\mathbb{R}^d$ whatsoever, lattice or not, beats weight 1, 4, 8, 16; and $|E| \leq K(d)$ is attained for $d = 2, 3, 4$. Given $|S| = 496$, the values 197 056, 198 550, 200 540, 204 520 are exactly what this template yields, and improving them requires a larger S . For $d = 5, 6, 7$ the kissing numbers are unknown ($K(5) \geq 40$, $K(6) \geq 72$, $K(7) \geq 126$), so $n = 29, 30, 31$ could still move — but only if one of those small-dimensional records moves. The partitions of D_5 , E_6 , E_7 themselves are optimal.

(b) *The case $d = 5$.* Here $K = 40 \equiv 1 \pmod{3}$, so the weight is at most 26, which [7] attain with twelve triangles and two pairs; the counting argument alone settles optimality. A sharper fact explains why one cannot instead “round up” to thirteen triangles.

Proposition 6. *Let T be a finite set of vectors of common squared norm $N > 0$ with $\sum_{y \in T} y = 0$ (for instance a root system). Then every equilateral triangle $\{x, y, z\} \subseteq T$ satisfies $x + y + z = 0$; and if $|T| \equiv 1 \pmod{3}$, then T contains no family of $(|T| - 1)/3$ pairwise disjoint equilateral triangles.*

Proof. The pairwise inner products in a triangle are $-N/2$, so $|x + y + z|^2 = 3N + 6(-N/2) = 0$. Such a family would cover all but one point p ; summing, $p = \sum_{y \in T} y - \sum_{\text{triangles}} (x + y + z) = 0$, contradicting $|p|^2 = N > 0$. $\square$

For D_5 , $|T| = 40 = 3 \cdot 13 + 1$, so at most 12 disjoint triangles fit — the twelve that [7] use. (Thirteen triangles would have weight 26 anyway, so Proposition 6 refines the optimality claim rather than being needed for it.) The argument applies verbatim to any antipodal configuration of size $\equiv 1 \pmod{3}$.

5. THE CONFIGURATION AND ITS VERIFICATION

Since the claim is a record, we state exactly what is checked. The accompanying data are: `S_01.txt–S_04.txt`, the four sets S_i , one vector per line as 24 integers of squared norm 32 (integer coordinates scaled by $\sqrt{8}$ relative to Section 2, so that the “no 60° ” condition reads $\langle x, x' \rangle \leq 8$ and the Leech minimum is 32); `T.txt`, the twelve cuboctahedron vectors in the D_3 model with the four triangles; `extra.txt`, the twelve extra spheres in exact form $y' = (a + b\sqrt{2})/2$, $a, b \in \mathbb{Z}^3$; `family.json`, the same data machine-readably with SHA-256 digests; and `verify.py`, a self-contained checker requiring only Python 3 and NumPy.

The checker rebuilds the extended binary Golay code from $g(x) = x^{11} + x^{10} + x^6 + x^5 + x^4 + x^2 + 1$ and confirms its weight distribution $(1, 759, 2576, 759, 1)$; regenerates all 196 560 minimal vectors from the shapes $(\pm 2^8, 0^{16})$, $(\mp 3, \pm 1^{23})$, $(\pm 4, \pm 4, 0^{22})$ and checks each against the arithmetic membership test for Λ_{24} and the inner-product histogram $(1, 4600, 47104, 93150, 47104, 4600, 1)$; and then verifies, in exact integer arithmetic, that each S_i lies in C with 496 distinct elements and off-diagonal Gram entries ≤ 8 , that the four sets are pairwise disjoint, that the twelve T -vectors have pairwise cosine $\leq 1/2$ with the four groups disjoint and internally at cosine exactly $-1/2$, that the twelve extra spheres (whose coordinates lie in $\frac{1}{2}\mathbb{Z}[\sqrt{2}]$) satisfy the two angle conditions — decided by sign tests on integers $A + B\sqrt{2}$, never in floating point — and that the count is $12 + 196560 + 8 \cdot 496 = 200540$.

Every one of the eight pair classes of Theorem 2 is then checked, with the class sizes summing to

$$\binom{200540}{2} = 20\,108\,045\,530,$$

so that no pair is left unexamined. The equatorial–equatorial class is covered by the Leech minimum ($x \neq x'$ in C gives $x - x' \in \Lambda_{24} \setminus \{0\}$, hence $|x - x'|^2 \geq 32$ and $\langle x, x' \rangle \leq 16$ in integer coordinates); the flag `--full` replaces this by a brute-force pass over all 19 317 818 520 pairs of C , confirming the maximum 16. The flag `--float` additionally assembles the 200540 explicit rows in $\mathbb{R}^{27}$ and computes the maximum off-diagonal inner product by a tiled matrix product, obtaining exactly 2.000000000000 — attained on lifted–lifted pairs sharing the same x , where the bound is tight ($\frac{8}{3} - \frac{2}{3} = 2$), and also on lifted–lifted pairs across sets with $\langle x, x' \rangle = 2$ and $\langle y, y' \rangle = 1/2$. On an eight-core laptop the three passes take about two seconds, ninety seconds and a few minutes respectively; all were run and all pass.

The configuration was also verified by two independently written programs: a Python verifier for the general template of Theorem 2 and a C++ checker of the set-level conditions. Both report 200540 and maximum inner product 2. Finally, the certificate is not tied to [7]’s particular sets: four pairwise disjoint 496-element sets follow from any single one by taking $g \in \text{Co}_0 = \text{Aut}(\Lambda_{24})$ with $g^i S \cap S = \emptyset$ for $i = 1, 2, 3$ and setting $S_i = g^{i-1} S$. Empirically a random g satisfies $gS \cap S = \emptyset$ about 54% of the time, so a few trials suffice; a family obtained this way was checked to give 200540 as well.

Data availability. The certificate files and the checker accompany this note. The four 496-element sets derive from the data of [7] (<https://github.com/CDM1619/PackingStar>); their provenance — source file, SHA-256 digest, and the coordinate map used — is recorded in the headers of the set files.

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Email address: benlindow@gmail.com